Year 13 Further Maths - Chi-squared tests

Similar questions - Contingency tables and goodness of fit

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 Maya investigates whether being left-handed is associated with living in a particular town. She obtains data from random samples from towns A and B; L denotes "left-handed".

	L	L'
A	18	12
B	25	45

(a) Test at the 10% significance level whether there is association between being left-handed and the town. **[7]**

(b) A person is chosen at random from one of the two towns. Let A denote "the person lives in town A". State the relationship between $\mathrm{P}(L)$ and $\mathrm{P}(L\,|\,A)$ implied by the null hypothesis. **[1]**

(c) Use the data to suggest a value for $\mathrm{P}(L\,|\,A)$ for an improved model. **[2]**

2 A manager believes that on average the numbers of absences on Monday to Friday are in the ratio 3 : 2 : 2 : 2 : 3. In a random sample of 120 absences the numbers were 33, 17, 21, 14 and 35 respectively.

(a) Test at the 5% significance level whether the data are consistent with the manager's belief. **[7]**

(b) A second independent random sample of size n gives integer numbers of absences in exactly the same ratio as the first sample. Find the smallest value of n for which the conclusion of the test is that the data are *not* consistent with the manager's belief. **[3]**

3 A coach records the times X minutes, of 300 runners in a race, and tests at the 5% level whether $X \sim \mathrm{N}(50, 8^2)$. The grouped data are:

Time	$X < 40$	$40 \leqslant X < 48$	$48 \leqslant X < 52$	$52 \leqslant X < 60$	$X \geqslant 60$
Observed	18	70	62	100	50

(a) State suitable hypotheses. **[1]**

(b) Show that the expected frequency for $48 \leqslant X < 52$ is 59.22 (2 d.p.). **[3]**

(c) Carry out the test. **[4]**

(d) Suggest one feature of the data that an improved model should take into account. **[1]**

4 A random sample of 80 students completed a task in a time T (as a fraction of the maximum time, so $0 \leqslant T \leqslant 1$) with $\sum t = 44$. A researcher proposes a probability density function $\mathrm{f}(t) = \alpha t^{\alpha-1}$ for $0 \leqslant t \leqslant 1$ ($\alpha > 0$).

(a) **In this part you must show detailed reasoning.** By finding $\mathrm{E}(T)$ and using the sample mean, determine an estimate of α as an exact fraction. **[6]**

The numbers of students with T in $[0, \frac{1}{3})$, $[\frac{1}{3}, \frac{2}{3})$ and $[\frac{2}{3}, 1]$ were 20, 28 and 32.

(b) Using the value of α in part (a), determine the extent to which the model is a good one (do not carry out a goodness of fit test). **[4]**

Question 1

(a) H_0: no association between handedness and town; H_1: there is association. Expected: A: L 12.9, L' 17.1; B: L 30.1, L' 39.9. $\chi^2 = \sum \dfrac{(O-E)^2}{E} = 5.05$. Degrees of freedom 1; critical value 2.706. $5.05 > 2.706$ so reject H_0: there is evidence of association (at the 10% level)

(b) $P(L) = P(L|A)$ (independence)

(c) $P(L|A) \approx \dfrac{18}{30} = 0.6$

Question 2

(a) H_0: absences are in the ratio $3:2:2:2:3$; H_1: not. Expected $30, 20, 20, 20, 30$; contributions $0.3, 0.45, 0.05, 1.8, 0.833$; $\chi^2 = 3.43$, critical value ($\nu = 4$) 9.488. $3.43 < 9.488$ so accept H_0: consistent with the belief

(b) Counts are proportional, so χ^2 scales with n: $\chi^2 = 3.43 \times \dfrac{n}{120} > 9.488 \Rightarrow n > 331.7$. Integer counts in the ratio $33:17:21:14:35$ need n to be a multiple of 120 (since gcd $= 1$ and the total is 120), so the smallest is $n = 360$

Question 3

(a) H_0: $X \sim N(50, 8^2)$; H_1: it is not

(b) $P(48 \leqslant X < 52) = P(-0.25 \leqslant Z < 0.25) = 0.1974$; $300 \times 0.1974 = 59.22$

(c) Expected: $31.7, 88.7, 59.2, 88.7, 31.7$; contributions: $5.92, 3.94, 0.13, 1.44, 10.57$; $\chi^2 = 22.00$; $\nu = 5 - 1 = 4$; critical value 9.488. $22.00 > 9.488$ so reject H_0 (the normal model is not suitable)

(d) Any sensible feature, for example the data are skewed: far more runners finish well above 60 than below 40 (observed 50 and 18), so a skewed model is needed

Question 4

(a) $E(T) = \int_0^1 t \cdot \alpha t^{\alpha-1} dt = \dfrac{\alpha}{\alpha+1}$. Sample mean $= \dfrac{44}{80} = 0.55$, so $\dfrac{\alpha}{\alpha+1} = \dfrac{11}{20}$, $20\alpha = 11\alpha + 11$, $\alpha = \dfrac{11}{9}$

(b) $P(T < \frac{1}{3}) = \left(\frac{1}{3}\right)^{\alpha} = 0.2611$, $P(T < \frac{2}{3}) = \left(\frac{2}{3}\right)^{\alpha} = 0.6092$. Expected frequencies: 20.9, 27.8, 31.3 against observed 20, 28, 32. These agree closely (all within about 1), so the model with $\alpha = \frac{11}{9}$ fits well